

Experiment No 11.

INTERFACING BLUETOOTH SENSOR WITH RASPBERRY PI

AIM: To study interfacing of Bluetooth with Raspberry pi.

REQUIREMENTS:- Raspberry pi board, 40 pin FRC cable, 4 pin female to female connector, 5v,2A power supply.

PROCEDURE:

- 1) Connect Raspberry pi board.
- 2) Connect the HDMI screen, mouse and keyboard to Raspberry pi .
- 3) Switch ON the power supply.
- 4) Connect 40 pin FRC cable from to the 40 pin connector which is placed on the board. And connect pin female to female connector from the board J6 connector to “RGB” section.
- 5) Need to install bluedot libraries, BlueDot allows you to control your projects wirelessly
 - Open the Terminal window of and Install the Python library:

sudo pip3 install bluedot

To upgrade to the latest version:

sudo pip3 install bluedot --upgrade

- Get the Android Blue Dot app or use the Python Blue Dot app on your Android Phone.

Note : You don't need to install the library all the time, once library is installed you only need to pair your android phone to with Bluetooth.

- 6) Pair a and Android phone

On your Android phone:

1. Open Settings
2. Select Bluetooth and make your phone “discoverable”

On your :

1. Click Bluetooth ▸ Turn On Bluetooth (if it's off)

2. Click Bluetooth ▸ Make Discoverable
3. Click Bluetooth ▸ Add Device
4. Your phone will appear in the list, select it and click **Pair**

On your Android phone and .

1. Confirm the pairing code matches
2. Click OK

7) **Using the Command Line**

On your Android phone:

1. Open Settings
2. Select Bluetooth and make your phone “discoverable”

On your :

1. Open terminal and type “**sudo bluetoothctl**” and press Enter to open Bluetooth control
2. At the [bluetooth]# prompt enter the following commands:

discoverable on

pairable on

agent on

default-agent

scan on

3. Wait for a message to appear showing the Android phone has been found:
4. [NEW] Device 12:23:34:45:56:67 (MAC ADDRESS) devicename
5. Type pair with the mac address of your Android phone:
6. pair (MAC ADDRESS)

On your Android phone and .

1. Confirm the passcode.
2. Type **quit** and press Enter to return to the command line

8) **Connecting**

9) Run commands through terminals

10) Open the Terminal window of and enter the command >>
cd Documents/IOT_SHALA

Now we are into the folder **/ IOT_SHALA**

To see the list of files into **/ IOT_SHALA** folder enter command **>> ls**

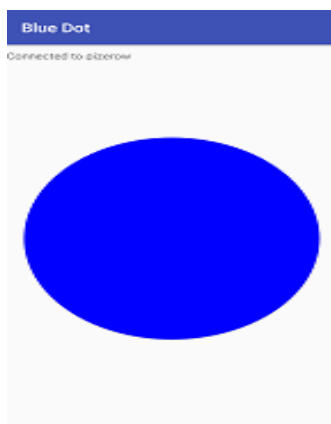
To run this specific program enter the command **>>**

python3 Test_Bluetooth_Button.py

- 11) To exit from the program press **Ctrl-Z**.
- 12) Start-up the Blue Dot app on your Android phone
 - Select your from the list



- Press the Blue Dot



Note

Your python program will need to be running and Waiting for connection before the BlueDot app will be able to connect to your